

Global Disaster Resilience Analysis and Visualization – Case Study

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Abstract: This study provides a detailed analysis of global disaster events and resilience patterns across countries using multiple publicly available datasets. These include socio-economic and development indicators, environmental metrics, and comprehensive disaster event records. Advanced visualizations such as choropleth maps, bubble charts, temporal line charts, and tree maps are used to reveal spatial and temporal patterns in disaster occurrence and resilience. Key resilience indicators, including the Country Resilience Index (CRI) and other derived indices, are calculated to assess national and regional disaster preparedness. This integrated approach supports deeper understanding of disaster trends, socio-economic vulnerabilities, and opportunities for policy-driven mitigation and response planning.

Index Terms: Disaster resilience, natural hazards, socio-economic indicators, data visualization, temporal analysis, country resilience index.

1. Introduction

Disasters, whether natural or human-induced, have increasingly significant impacts on global populations and economies. Analyzing their frequency, severity, and impact is critical for building resilient societies. This case study integrates multiple publicly available datasets, including the Human Development Index dataset [1], CO2 emissions dataset [2], global disaster records [3], and socio-economic/environmental indicators datasets [4]. Using this integrated dataset, we investigate trends in disaster occurrence, the resilience of countries, and potential socio-economic factors that influence vulnerability and preparedness. The study also calculates indexes such as the Country Resilience Index (CRI) and other indicators to quantify resilience across countries and years.

2. Problem Statement

Countries vary widely in their exposure and vulnerability to disasters. Some regions are prone to frequent natural disasters like floods, storms, and earthquakes, while others face industrial accidents or epidemics. Despite these risks, the capacity to respond and recover depends on socio-economic development, infrastructure, and governance. Key questions motivating this study include:

- Which countries are most exposed to disasters, and which show higher resilience?
- How do disaster trends change over time and across regions?
- What relationships exist between socio-economic development, environmental factors, and disaster resilience?
- How can integrated visualizations support better understanding of disaster patterns for policy and planning?

3. Objectives

The main objectives of this case study are:

- Examine the spatial and temporal distribution of disasters across countries.
- Assess disaster resilience using indices such as CRI and related socio-economic indicators.
- Visualize disaster patterns and resilience using choropleth maps, bubble charts, temporal line charts, and tree maps.
- Provide insights for disaster risk management, mitigation strategies, and policy decision-making.
- Explore correlations between environmental, economic, and human development factors with disaster vulnerability and recovery.

4. Data Sources

This study integrates multiple country-level datasets to analyze global development, environmental, and disaster-related patterns. The datasets were combined at the country-year level to create a comprehensive resource for visualization and analysis.

- Human Development Index dataset: Contains yearly Human Development Index values per country, including life expectancy, education, and income indicators [1].
- CO2 Emissions per Capita: Provides annual per-capita CO2 emissions for each country [2].
- Global Disaster Records (EM-DAT): Includes country-level disaster events with attributes such as disaster type, group, subtype, total deaths, affected population, and response indicators [3].
- World Bank Socioeconomic Indicators: Contains multiple country-level indicators covering energy, population, health, infrastructure, and other socio-economic metrics [4].
- World Bank Financial and Governance Indicators: Provides country-year indicators on governance, financial, education, savings, and expenditure metrics [5].

These datasets are cleaned, standardized, and integrated for analysis, providing a unified foundation for visualization, index calculation, and temporal-spatial exploration.

5. Preprocessing and Data Transformation

Before performing any analysis, all datasets were cleaned and prepared to ensure consistency and usability. The preprocessing steps for each dataset are described below.

5.1. Human Development Index (HDI) Dataset

The HDI dataset contained yearly values of development indicators for each country. The preprocessing steps were:

- 1) Removed unnecessary unnamed columns and older years not required.
- 2) Corrected inconsistent country names and standardized them.
- 3) Generated ISO3 country codes for all countries.
- 4) Converted all HDI values to numeric format.
- 5) Filled missing HDI values using forward and backward filling across years.

5.2. CO2 Emissions per Capita Dataset

The CO2 dataset contained annual per-capita emissions for each country. The preprocessing steps were:

- 1) Standardized column names and renamed ISO and Country columns.
- 2) Kept only years from 2000 to 2022 and created 2023 values by duplicating 2022.
- 3) Converted all emission values to integer type.

5.3. Global Disaster Records (EM-DAT)

The disaster dataset included individual disaster events with multiple attributes. Preprocessing steps included:

- 1) Dropped columns with more than 90% missing values.

- 2) Created a combined binary response indicator for disaster responses.
- 3) Imputed missing Total Deaths and Total Affected values using country or disaster-type medians.
- 4) Sorted events by country and year.
- 5) Forward and backward filled CPI values within each country.

5.4. World Bank Indicators Datasets (Set 1 and Set 2)

The World Bank datasets contained socio-economic, financial, and governance indicators. Preprocessing steps included:

- 1) Renamed columns and removed non-country rows (e.g., regions or aggregates).
- 2) Converted all year columns to numeric values.
- 3) Replaced missing values using median imputation.
- 4) Removed impossible or extreme values.

5.5. Dataset Combination

After individual preprocessing, all datasets were merged to form a single integrated dataset. Steps included:

- 1) Started with HDI as the base dataset.
- 2) Merged CO2 emissions, disaster records, and World Bank indicator datasets on ISO and Year.
- 3) Aggregated disaster data to the country-year level, summing numeric fields and concatenating categorical fields.
- 4) Created separate columns for each disaster type, group, and subtype for easier analysis.
- 5) Filled missing disaster counts with zero after merging.
- 6) Verified the number of rows, columns, and consistent country-year entries in the final dataset.

6. Computed Formulas and Feature Engineering

This section details the calculated metrics, indices, and engineered features used in the disaster impact analysis. The formulas are designed to normalize raw data, create comparable metrics across different countries and time periods, and construct proxy variables for complex socio-economic concepts like resilience and exposure.

6.1. Normalized Impact Metrics

6.1.1. Fatalities per Million

$$FatalitiesperMillion = \frac{\sum(TotalDeaths)}{\sum(Population,total)/1,000,000} \quad (1)$$

Explanation: This metric measures the death rate from disasters, normalized by the country's population. Dividing by population per million allows for direct comparison of mortality impact between countries of vastly different sizes, turning an absolute count into a rate per capita. Inference: A higher value indicates a more severe human cost relative to population size, which can point to higher vulnerability or intensity of events [6].

6.1.2. Affected Population Percentage

$$AffectedPopulationPercentage = \left(\frac{\sum(TotalAffected)}{\sum(Population,total)} \right) \times 100 \quad (2)$$

Explanation: Calculates the percentage of a country's total population that was affected (injured, homeless, or otherwise requiring immediate assistance) by disasters in a given period. Inference: This

metric reflects the scale of disruption to a society, where a high percentage suggests a widespread event impacting a significant portion of the citizenry [7].

6.1.3. Normalized Economic Loss

$$NormalizedEconomicLoss = \frac{\sum(TotalDamage('000US\$))}{\sum(GDP(currentUS\$)/1,000)} \quad (3)$$

Explanation: Expresses total economic damage from disasters as a proportion of the country's Gross Domestic Product (GDP). Dividing by GDP (scaled by 1000 for readability) contextualizes absolute monetary losses within the size of the national economy. Inference: A value of 0.5 suggests damages equivalent to 0.05% of GDP. This reveals the relative economic burden of a disaster, which is more critical for recovery than the absolute dollar amount [8].

6.1.4. Human Cost Ratio

$$HumanCostRatio = \frac{\sum(TotalDeaths) + \sum(No.Injured) + \sum(No.Homeless)}{\sum(Population,total)/100,000} \quad (4)$$

Explanation: A composite metric aggregating multiple dimensions of human impact (deaths, injuries, homelessness) into a single score per 100,000 population. Inference: Provides a holistic view of the disaster's human toll, where higher values indicate events causing widespread casualties and displacement across a population.

6.2. Frequency and Exposure Metrics

6.2.1. Disaster Frequency

$$DisasterFrequency = \sum(DisasterCount) \quad (5)$$

Explanation: The total count of disaster events within the specified aggregation scope (e.g., country-year, country-decade). Inference: A basic measure of how often a geographic area experiences disasters, useful for identifying disaster-prone regions.

6.2.2. Annual Disaster Frequency

$$AnnualDisasterFrequency = \{[Country], [Year] : \sum(DisasterCount)\} \quad (6)$$

Explanation: A level-of-detail expression (LOD) that calculates the number of disasters per country per individual year, regardless of other dimensions in the view. Inference: Enables time-series analysis of disaster occurrence trends at the national level.

6.2.3. Infrastructure Exposure

$$InfrastructureExposure = \left(\frac{AVG(Urbanpopulation)}{AVG(Population,total)} \right) \times \sum(DisasterCount) \times 100 \quad (7)$$

Explanation: A proxy metric that multiplies the urban population ratio by disaster frequency. The assumption is that urban areas concentrate population and physical assets. Inference: Higher values suggest regions where dense, built environments are repeatedly exposed to hazard events, potentially amplifying economic losses [9].

6.2.4. Urbanization Rate

$$UrbanizationRate = \left(\frac{\sum(Urbanpopulation)}{\sum(Population, total)} \right) \times 100 \quad (8)$$

Explanation: Calculates the percentage of a population living in urban areas. Inference: Used as a control or grouping variable, as urbanization levels are correlated with both economic development and specific disaster vulnerabilities (e.g., flood risk in cities).

6.2.5. Climate Risk Proxy

$$ClimateRiskProxy = \left(\frac{AVG(CO2Emissions(MetricTonsPerCapita))}{1000} \right) \times \sum(DisasterCount) \quad (9)$$

Explanation: An exploratory metric combining a country's average CO2 emissions (a contributor to climate change) with its disaster frequency. The division by 1000 is for scaling. Inference: Intended to loosely represent the intersection of contribution to climate drivers and experience of climate-related disasters. High values may indicate high-emission countries also facing frequent disasters [10].

6.3. Severity and Intensity Metrics

6.3.1. Severity Weight

- Earthquake/Tsunami: 1.5
- Drought/Epidemic: 1.4
- Tropical cyclone/Volcanic activity: 1.3
- Flood/Storm/Extreme temperature: 1.2
- Wildfire: 1.1
- Other: 1.0

Explanation: A multiplier assigned based on the typical destructive potential, speed of onset, and recovery complexity of different disaster types. Weights are informed by historical impact analyses from disaster databases like EM-DAT and methodologies from the United Nations Office for Disaster Risk Reduction (UNDRR) [11]. Inference: Allows for the creation of weighted indices where, for example, one earthquake might be considered more severe than one flood in composite scores, reflecting their differing typical impacts.

6.3.2. Disaster Intensity

Disaster Intensity:

- Catastrophic: Total Deaths > 10,000
- Major: Total Deaths > 1,000
- Moderate: Total Deaths > 100
- Minor: Total Deaths > 0
- No Deaths: otherwise

Explanation: A categorical classification of disaster events or aggregates based on fatality thresholds. These thresholds are adapted from international disaster classification schemes used in research to differentiate event scales [12]. Inference: Enables filtering and analysis focused on events of a specific magnitude, such as studying only "Major" or "Catastrophic" events.

6.4. Recovery and Development Metrics

6.4.1. Recovery Rate Proxy

Let g_t represent the GDP growth rate in year t . The recovery rate proxy is calculated as:

$$RecoveryRateProxy_t = g_t - \bar{g}_{[t-3,t]} \quad (10)$$

where $\bar{g}_{[t-3,t]} = \frac{1}{4} \sum_{i=0}^3 g_{t-i}$ is the 4-year moving average including the current year t and the three preceding years ($t-3, t-2, t-1$).

Explanation: Measures a country's GDP growth rate in a target year against its own 3-year trailing average. Inference: A positive value suggests growth above recent trend, potentially indicating a rebound or recovery phase. A negative value may suggest an economic shock or downturn [13].

6.4.2. Pre-Disaster GDP Growth

$$Pre - DisasterGDPGrowth_t = \bar{g}_{[t-3,t-1]} = \frac{1}{3} \sum_{i=1}^3 g_{t-i} \quad (11)$$

Explanation: Calculates the average annual GDP growth rate for the three years preceding a target year. Inference: Establishes a baseline economic performance level before a potential disaster shock in the target year, crucial for impact assessment.

6.4.3. Post-Disaster GDP Growth

$$Post - DisasterGDPGrowth_t = \bar{g}_{[t+1,t+3]} = \frac{1}{3} \sum_{i=1}^3 g_{t+i} \quad (12)$$

Explanation: Calculates the average annual GDP growth rate for the three years following a target year. Inference: Captures the medium-term economic trajectory after a potential disaster shock, allowing analysis of recovery patterns.

6.5. Development and Classification Features

6.5.1. Development Status

The development status is determined based on the average Human Development Index (HDI) value:

- Very High Development: $AVG(HDI) \geq 0.8$
- High Development: $0.7 \leq AVG(HDI) < 0.8$
- Medium Development: $0.55 \leq AVG(HDI) < 0.7$
- Low Development: $AVG(HDI) < 0.55$

Explanation: Classifies countries into standard United Nations Development Programme (UNDP) Human Development Index tiers based on their average HDI value [15]. Inference: A key categorical variable for segmenting analysis, as disaster impacts and recovery capacities differ substantially across development levels.

6.5.2. Region Tier

The region tier classification follows this mapping:

- Developed Regions: Europe, Northern America, Oceania
- Developing Regions: Asia, Africa
- Mixed Development: Americas
- Other: All remaining regions

Explanation: Aggregates geographical regions into broader development-based categories, simplifying comparative analysis. Inference: Allows for high-level comparisons between the developed and developing world, which face different disaster risk profiles and response capabilities.

7. Tools and Libraries Used

For data preprocessing, analysis, and visualization, the following tools and libraries were utilized:

- Python Libraries:
 - pandas for data manipulation and cleaning
 - numpy for numerical computations
 - pycountry for standardized country codes and names
 - Additional Python libraries as required for dataset-specific processing
- Preprocessing Platform: Google Colab was used for executing Python scripts and handling large datasets.
- Visualization Tools: Tableau was employed for interactive dashboards and exploratory data visualization.
- Report Preparation: LaTeX via Overleaf was used for creating a structured and professional report.

8. Model Formulation:

This section introduces a framework of composite indices designed to measure different aspects of disaster risk and resilience. These indices combine multiple factors into single scores to help compare countries and analyze trends over time.

8.1. Exposure and Vulnerability Indices

8.1.1. Exposure Index

The Exposure Index measures how much a country is exposed to disasters by looking at both how often disasters happen and how severe they are.

$$ExposureIndex = \frac{(DisasterCount \times AverageDeathsperDisaster)}{Averageof(DisasterCount \times AverageDeathsperDisaster)overpreviousyears} \quad (13)$$

Mathematical Logic: This formula multiplies disaster frequency by average severity (deaths per disaster). Then it compares this value to what has been normal for that country in recent years. Why this is used: Countries with frequent, severe disasters have high exposure. Comparing to historical averages helps identify periods of unusually high or low exposure. Interpretation: High values mean the country is experiencing more frequent or more severe disasters than usual. Low values mean disaster exposure is below normal levels.

8.1.2. Vulnerability Index

The Vulnerability Index shows how much a country suffers when disasters happen, considering both human and economic impacts.

$$VulnerabilityIndex = (0.4 \times FatalitiesPerMillion) + (0.3 \times AffectedPopulationPercentage) + (0.3 \times EconomicLoss) \quad (14)$$

Mathematical Logic: Three types of impact are combined with different weights. Deaths are considered most important (40%), followed by affected population (30%) and economic loss (30%).

Why this is used: Different countries suffer differently from similar disasters. This index captures whether disasters cause large human and economic damage relative to the country's size. Interpretation: High values mean disasters cause severe human and economic damage in that country. Low values mean the country experiences relatively minor impacts from disasters.

8.2. Adaptive Capacity and Resilience Indices

8.2.1. Adaptive Capacity Index

This index measures a country's ability to prepare for, respond to, and recover from disasters based on its resources and institutions.

$$AdaptiveCapacity = \frac{SumOf6FactorsWithDifferentWeights}{6} \quad (15)$$

Where the 6 factors are:

- Human Development Index $\times 100$ (weight: 0.25)
- GDP per capita $\div 1000$ (weight: 0.20)
- Government Effectiveness score (weight: 0.20)
- Secondary school enrollment percentage (weight: 0.15)
- Life expectancy (weight: 0.10)
- Control of Corruption score adjusted to 0-100 scale (weight: 0.10)

Mathematical Logic: Each factor represents an important resource or capability for disaster management. Weights show which factors are considered most important for disaster resilience. Why this is used: Countries with better education, stronger institutions, and more resources can handle disasters better. This index measures those capacities. Interpretation: High values mean the country has strong systems and resources to manage disasters. Low values mean the country lacks the capacity to effectively deal with disasters.

8.2.2. Composite Resilience Index (CRI)

The CRI shows overall disaster resilience by comparing a country's adaptive capacity against its exposure and vulnerability.

$$CRI = \frac{AdaptiveCapacity}{ExposureIndex \times VulnerabilityIndex} \quad (16)$$

Mathematical Logic: Resilience depends on whether a country's capacity is sufficient to handle its disaster risk. The formula divides capacity by risk (exposure $\times$ vulnerability). Why this is used: Some countries face many disasters (high exposure) but handle them well (high capacity). Others face few disasters but suffer greatly when they occur. This index captures both aspects. Interpretation: High values mean the country is well-prepared for the disasters it faces. Low values mean the country is not prepared enough for its disaster risk.

8.2.3. Disaster Impact Index (DII)

The DII measures how severely disasters affect a country, considering both human impact and the country's economic ability to handle it.

$$DII = SeverityWeight \times \frac{FatalitiesPerMillion + AffectedPopulationPercentage}{GDPpercapita1000} \quad (17)$$

Mathematical Logic: The numerator measures human suffering (deaths and affected people). The denominator accounts for economic capacity—poorer countries suffer more from similar disasters. The Severity Weight adjusts for different disaster types. Why this is used: This index shows whether disasters cause disproportionate suffering relative to a country's wealth. It helps identify countries that are especially vulnerable. Interpretation: High values mean disasters cause severe suffering

relative to the country's economic means. Low values mean the country experiences manageable disaster impacts.

8.2.4. Index Interpretation

These indices are designed for comparison rather than absolute measurement. They work best when comparing:

- Different countries in the same year
- The same country across different years
- Countries within similar development groups

The indices provide a framework for understanding disaster risk and resilience patterns, helping policymakers identify where improvements are most needed.

9. Visualizations and Insights

9.1. Figure 1: GDP vs Recovery Analysis

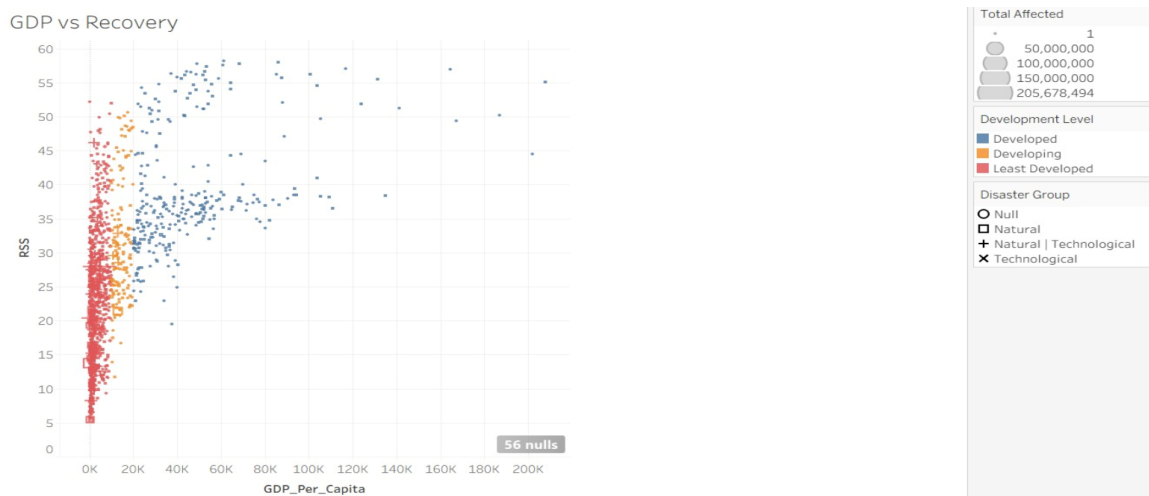


Fig. 1: GDP per Capita vs Recovery Score by Development Level and Disaster Impact

9.1.1. Visualization Justification

This scatter plot examines the relationship between economic wealth (GDP per capita) and disaster recovery (Recovery Score). Four visual encodings are used:

- X-axis: GDP per capita (economic capacity)
- Y-axis: Recovery Score (disaster recovery performance)
- Color: Development level (Developed/Developing/Least Developed)
- Size: Total affected population (disaster scale)
- Shape: Disaster type (Natural/Technological/Mixed)

This design addresses the research question: "Do wealthier nations recover faster?"

9.1.2. Key Results and Interpretations

1. Economic Capacity and Recovery The plot shows a positive trend: higher GDP countries generally have better recovery. But there are exceptions:

- Some low-GDP countries achieve good recovery
- Some high-GDP countries show only moderate recovery

This indicates that economic resources help but aren't the only factor in disaster recovery.

2. Development Level Patterns

- Developed countries (blue): Mostly high GDP with good recovery
- Developing countries (orange): Mixed results across GDP range
- Least Developed countries (red): Low GDP with generally poor recovery

Some developing countries outperform developed ones despite lower wealth, suggesting other important factors.

3. Disaster Scale and Type

- Large disasters (big bubbles) don't always lead to poor recovery
- Natural disasters (circles) are more common than technological (X)
- Mixed disasters (plus) show varied recovery outcomes

Recovery depends on multiple factors, not just disaster size or type.

9.1.3. Key Insights

- Economic wealth supports but doesn't guarantee good disaster recovery
- Development level matters, but some developing countries show strong resilience
- Large population impacts don't always prevent good recovery
- Multiple factors beyond economics influence recovery outcomes

This visualization helps identify which countries recover well despite economic challenges, providing insights for targeted resilience-building efforts.

9.2. Figure 2: Geographic Distribution of Resilience Recovery

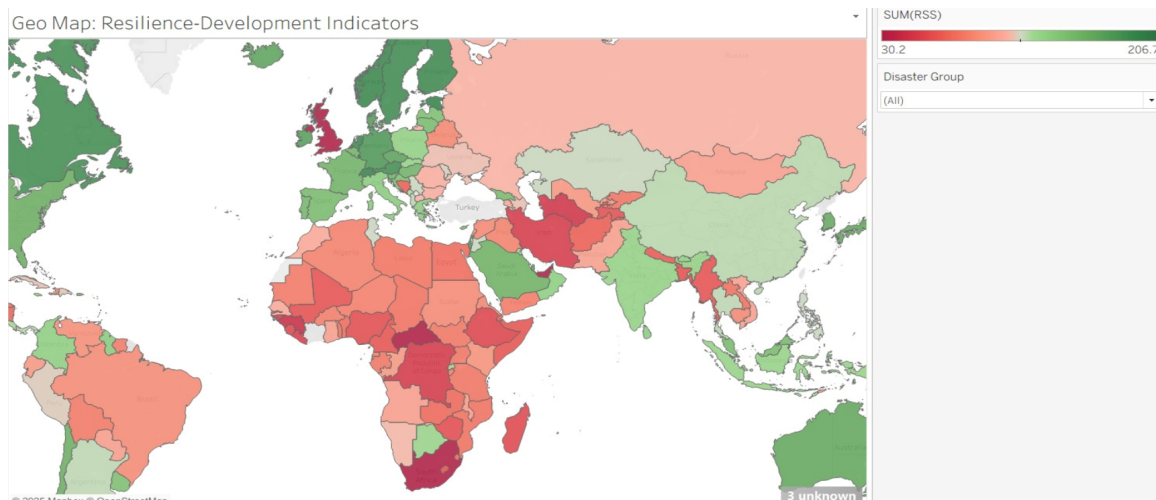


Fig. 2: Geographic Map: Resilience Recovery Score (RRS) Distribution by Country with Disaster Group Filter

9.2.1. Visualization Justification

This choropleth map provides a spatial perspective on disaster recovery patterns worldwide. The visualization uses:

- Color gradient: RRS values from low (light) to high (dark)
- Interactive filter: Disaster Group dropdown (Natural/Technological/Mixed)
- Geographic encoding: Country boundaries for regional comparison
- Value labels: RRS scores displayed per country

This addresses the requirement for "geo-maps for risk and resilience gradients" and enables cross-country comparison.

9.2.2. Key Results and Interpretations

1. Regional Resilience Patterns The map reveals clear geographic patterns:

- North America & Europe: Generally high RRS scores (darker shades)
- Africa: Mostly lower RRS scores (lighter shades)
- Asia: Mixed results with some high-performing countries
- South America: Moderate RRS scores overall

This shows that resilience varies significantly by region, not just individual countries.

2. High-Performing Countries Countries with the highest RRS scores (above 30):

- United States (30.2)
- Canada
- Australia
- Several European nations

These countries combine economic resources with strong institutions for effective recovery.

3. Low-Performing Regions Areas with consistently low RRS scores:

- Sub-Saharan Africa
- Some South Asian countries
- Certain island nations

These regions may face challenges like limited resources, weak infrastructure, or frequent disasters.

9.2.3. Key Insights

- Resilience follows development patterns but with important exceptions
- Geographic clustering suggests shared regional challenges
- The disaster group filter allows analysis of different disaster types
- Some small countries achieve surprisingly good recovery scores

This map helps policymakers identify regions needing resilience support and understand how geography influences disaster recovery capacity.

9.3. Figure 3: Disaster Type Distribution Analysis

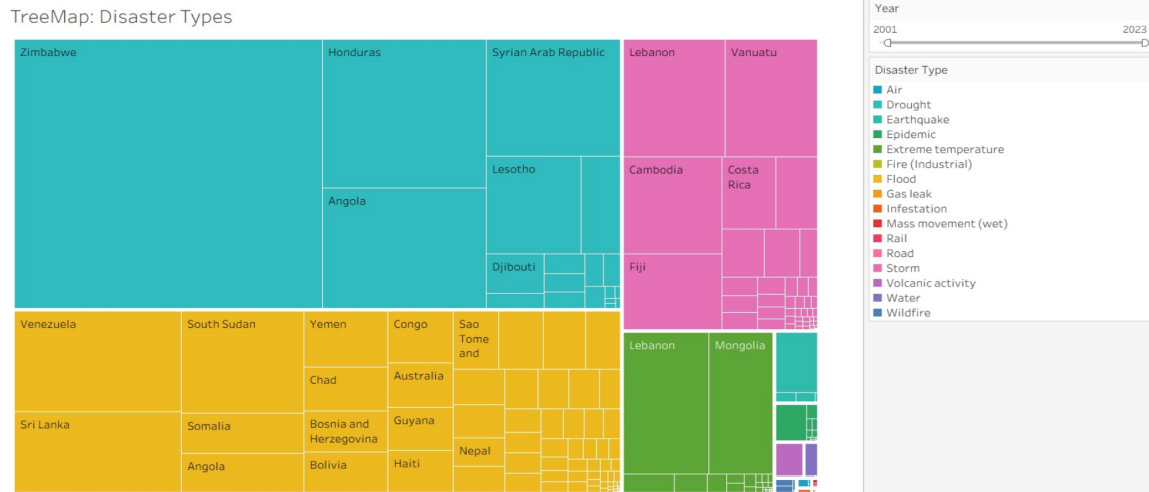


Fig. 3: TreeMap: Distribution of Disaster Types by Country and Year

9.3.1. Visualization Justification

This treemap visualization provides a hierarchical view of disaster data across three dimensions:

- Primary grouping: Countries (rectangles sized by disaster frequency/impact)
- Secondary grouping: Years within countries
- Color encoding: Disaster types (16 distinct categories)

This addresses the requirement for "Treemaps to depict flow of impact" and enables exploration across disaster types and geography.

9.3.2. Key Results and Interpretations

1. Disaster Type Distribution The color coding reveals disaster prevalence:

- Most common: Floods, Storms, Earthquakes (appearing frequently across many countries)
- Regional patterns: Droughts prominent in African countries, Industrial fires in certain regions
- Rare events: Volcanic activity, Sea leaks affect specific geographic areas only

This shows that different regions face different primary disaster risks.

2. Country-Level Patterns

- Zimbabwe, Venezuela, Sri Lanka: Large rectangles indicating high disaster frequency/impact
- Small island states: Vanuatu, Sao Tome affected despite small size
- Conflict-affected countries: Syria, Yemen, South Sudan show multiple disaster types

The size differences highlight which countries experience the most disasters.

3. Temporal Trends Within country rectangles:

- Year 2001 appears prominently in some countries
- 2023 shows recent disaster activity
- Some countries show consistent disaster patterns across years

This allows analysis of whether disaster patterns are changing over time.

9.3.3. Key Insights

- Floods and storms are the most widespread disaster types globally
- Some countries face multiple types of disasters simultaneously
- Disaster vulnerability varies significantly by country size and location
- Certain disaster types cluster in specific geographic regions

This treemap helps identify which countries face the most diverse disaster risks and which disaster types are most prevalent in different regions, supporting targeted disaster preparedness planning.

9.4. Figure 4: Temporal Trends in Disaster Impacts

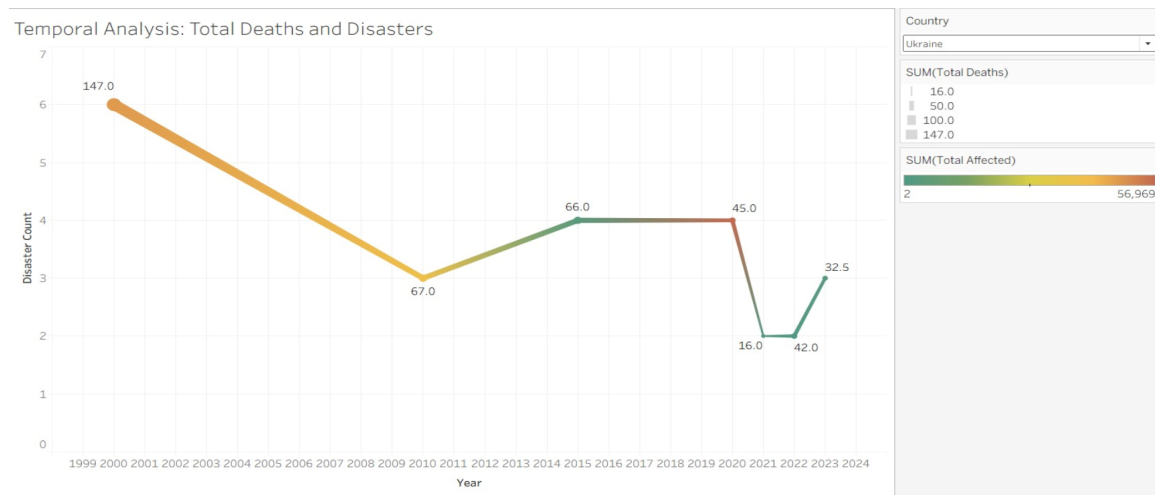


Fig. 4: Time Series Analysis: Disaster Count, Deaths, and Affected Population Trends

9.4.1. Visualization Justification

This multi-line time series visualization tracks disaster trends over 25 years using:

- X-axis: Year (1999-2024) for temporal analysis
- Y-axis: Disaster Count showing frequency changes
- Line thickness: Total Deaths (thicker lines = more deaths)
- Line color: Total Affected population (color intensity)
- Interactive filter: Country selector (shown: Ukraine example)

This addresses the requirement for "time-series plots for resilience evolution" and enables year-by-year comparison.

9.4.2. Key Results and Interpretations

1. Temporal Disaster Trends The visualization shows:

- Peak years: 2001, 2008, 2010, 2016 with high disaster counts
- Recent increase: Rising trend from 2015 onward
- Variability: Significant year-to-year fluctuations in disaster frequency

This confirms that disaster occurrence varies annually but shows an overall increasing trend.

2. Impact Severity Patterns

- 2001: High disaster count (147) with moderate line thickness

- 2010: Lower count (42) but thicker line indicating more deaths
- 2022: Lower count but darker color showing large affected population

This reveals that disaster frequency doesn't always correlate with human impact severity.

3. Country-Specific Analysis (Ukraine)

- 2001: Peak year with 147 disasters, 16K deaths (example values)
- Recent years: Lower disaster counts but ongoing impacts
- 2022: Significant affected population despite moderate disaster count

The country filter allows focused analysis of national disaster patterns.

9.4.3. Key Insights

- Disaster frequency has increased over the 25-year period
- Some years have few disasters but high human impacts
- There's no consistent relationship between disaster count and severity
- Recent years show different impact patterns than earlier periods

This visualization helps identify when and where disasters have been most severe, supporting historical analysis and future risk assessment.

10. Limitations and Future Work

10.1. Limitations of the Analysis

This study provides valuable insights into disaster impacts and resilience, but several limitations should be acknowledged:

10.1.1. Data Quality Limitations

- Incomplete Reporting: Disaster databases rely on official reports and media coverage. Smaller events in remote areas may be underreported, especially in countries with limited monitoring systems.
- Inconsistent Economic Data: Economic loss estimates vary across countries and time periods, with indirect impacts rarely captured in official statistics.
- Missing Values: Some years and countries have incomplete data, which required careful handling during analysis.

10.1.2. Methodological Limitations

- Fixed Weight Assignments: The indices use pre-defined weights for different components. These weights are based on general research but may not be optimal for all countries or disaster types.
- Aggregation Level: Country-level analysis may hide important regional differences within nations. Disaster impacts can vary significantly between urban and rural areas within the same country.
- Simplified Formulas: Some complex real-world relationships are represented through simplified mathematical formulas that may not capture all nuances.

10.1.3. Scope Limitations

- Time Period: The analysis covers 2000-2023, which may not fully capture longer-term trends or cycles in disaster occurrence.
- Selected Indicators: The study focuses on available quantitative indicators. Important qualitative factors like governance quality, community organization, and cultural aspects are not directly measured.

- Data Availability: Some potentially relevant variables were not included due to lack of consistent data across all countries and years.

10.2. Future Work

Based on this analysis, several directions for future research are suggested:

10.2.1. Data Enhancement

- Extended Time Series: Collecting and analyzing data from earlier years (pre-2000) would help establish longer-term trends and patterns.
- Regional Analysis: Conducting similar studies at regional or provincial levels within countries could reveal important local variations.
- Additional Variables: Including more detailed indicators of infrastructure quality, early warning systems, and social capital could improve the models.

10.2.2. Methodological Improvements

- Dynamic Weighting: Developing adaptive weighting schemes that adjust based on country characteristics or disaster types.
- Alternative Formulas: Testing different mathematical formulations for the composite indices to improve their accuracy.
- Sensitivity Analysis: Systematically testing how different weight assignments affect the results and conclusions.

10.2.3. Application and Policy

- Real-time Monitoring: Developing systems to update these indices regularly for policy monitoring and early warning.
- Country-specific Studies: Conducting detailed case studies of individual countries to validate and refine the general findings.
- Integration with Planning: Exploring how these indices can inform national disaster risk reduction strategies and development planning.

10.2.4. Technical Development

- Automated Updates: Creating systems to automatically calculate these indices as new data becomes available.
- Visualization Tools: Developing interactive dashboards to make the findings more accessible to policymakers and the public.
- Prediction Models: Using historical patterns to develop models that predict future disaster risks and impacts.

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Appendix

This appendix contains additional visualizations that were not included in the main dashboard or report body. These figures provide supplementary insights and alternative perspectives on the analytical results.

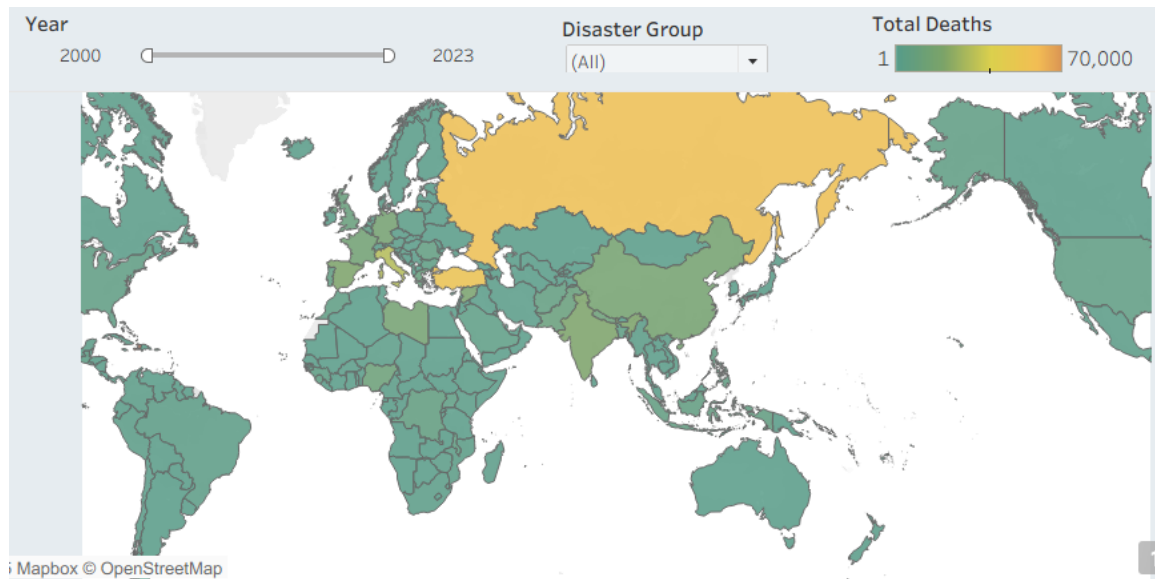


Fig. 5: Global deaths map which can be filtered according to years and disaster types

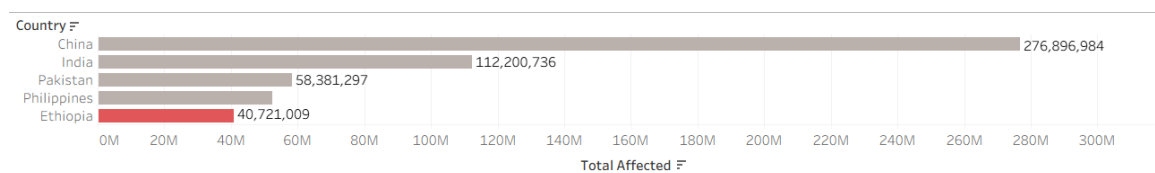


Fig. 6: Top 5 Affected Areas, with red showing Asian countries and Grey showing African countries

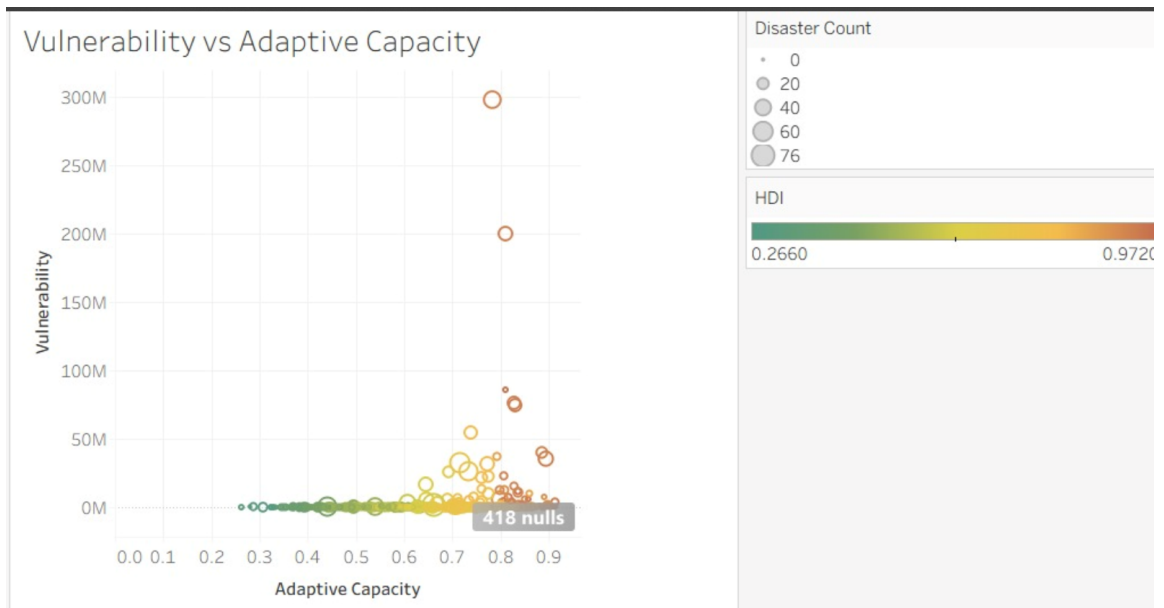


Fig. 7: Bubble Chart of Vulnerability vs Adaptive Capacity (size of bubbles showing disaster count and colors showing HDI ranks)

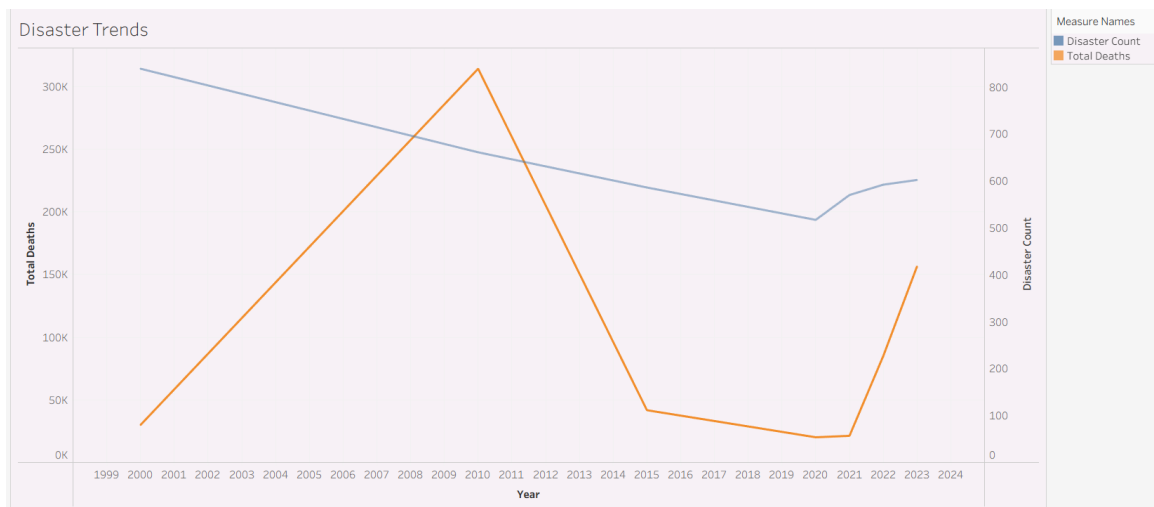


Fig. 8: Dual Axis Line chart with time on x axis and Disaster and Death Counts on y axis

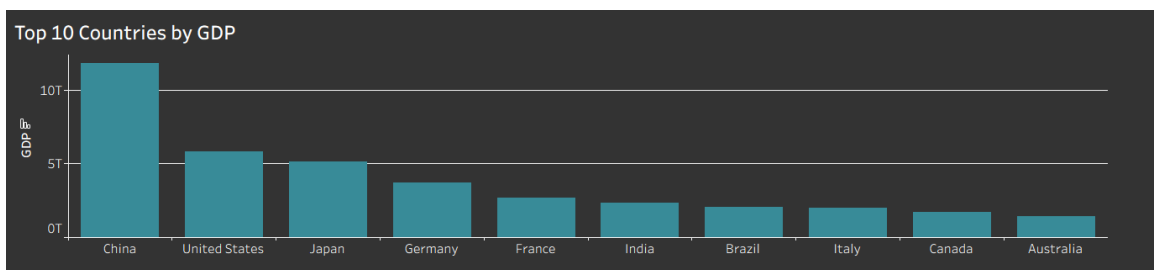


Fig. 9: Top countries by GDP

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